

# Conversational Graph Extraction

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## Headline Claim

Seo et al.'s structural metrics for schema quality, and LLM-simulated users seem to only track ground-truth utility within narrow conditions. This project maps where they hold and where they break.

## Research Questions and Scope

**RQ1.** Do Seo et al.'s six structural KG metrics predict downstream QA accuracy across schemas of varying quality? At least well enough to substitute for gold-ontology evaluation?

**RQ2.** Does feedback from an LLM-simulated user match feedback from human users, well enough to substitute simulated users for real users during development?

## What is Interesting

- Seo's six metrics measure graph shape. Whether shape predicts utility is an empirical question.
- Simulated users may be biased — a yes-man, an over-merger, an under-pruner — and the bias is invisible if you never compare to humans. They could also have prior knowledge that the normal human does not.

# Initial Thoughts on Experimentation

**Corpora (two, for cross-domain check):** still undecided on this.

## RQ1 plan:

- Produce schema variants per corpus by varying dialogue
- Score each schema with all six Seo metrics + downstream QA accuracy.
- Measure within and across domain.

## RQ2 plan:

- Get at least 5 humans (hopefully) to issue edits and then the simulated users to do the same, compare the responses
- Measure if schema's end up in the same place

# Initial System Architecture

| Component         | Role                                |
|-------------------|-------------------------------------|
| Corpus loader     | Loads + samples docs                |
| Schema store      | JSON schema                         |
| Corruption module | Induces RQ1 schema-quality variants |
| Metric            | Seo's six metrics + QA accuracy     |
| Logger            | JSON run records                    |

## Open Questions

- What counts as the "same domain" between two corpora?
- How to give a simulated user and a human equivalent access to the corpus, so the comparison is fair?
- How to score Seo metrics against any reference when no gold ontology exists?
- Is 300 docs per corpus enough to drive a measurable QA signal?

## State of the Art — What is Relevant

- **Zhang & Soh (2024)** — **EDC**. Three-phase LLM pipeline (extract, define, canonicalize). Reusable as my extraction backbone.
- **Pan et al. (2025)** — **SCHEMA-MINER**. Human-in-the-loop schema mining with iterative LLM + expert feedback. Direct ancestor of the conversational setting.
- **Kommineni et al. (2024)**. Uses an LLM-as-judge for evaluation — precedent for simulated feedback.
- **Seo et al. (2022)**. Six ground-truth-free structural KG metrics. Direct subject of RQ1.
- **Wang et al. (2024)**. Survey of KG evaluation metrics. Maps the gaps RQ1 addresses.